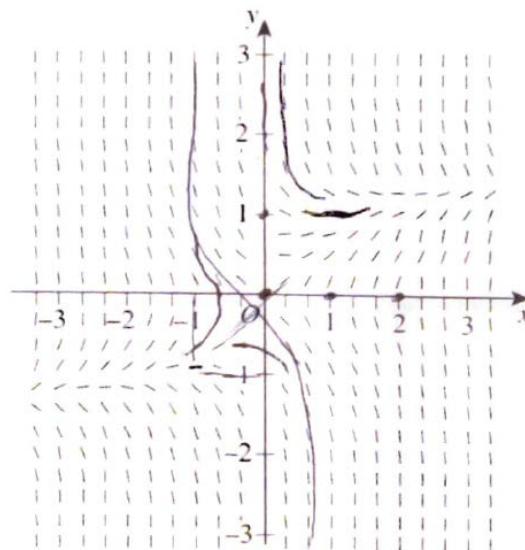


**Section I      10 marks****Attempt Questions 1-10****Allow about 15 minutes for this section**

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1. Which of the following is the angle between the vectors  $\vec{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$  and  $\vec{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$
- A.  $10^\circ$
- B.  $63^\circ$
- C.  $140^\circ$
- D.  $117^\circ$
2. Which of the following equal to  $\cos \theta$
- A.  $\sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2}$
- B.  $2 \cos^2 \frac{\theta}{2} + 1$
- C.  $1 - 2 \cos^2 \theta$
- D.  $\frac{\sin \theta}{\tan \theta}$
3. The polynomial  $P(x) = x^3 - x^2 - 5x - 3$  has a double root at  $x = \alpha$ . What is the value of  $\alpha$ ?
- A.  $-\frac{5}{3}$
- B.  $-1$
- C.  $1$
- D.  $\frac{5}{3}$
4. Which of the following is equivalent to  $\int \sin x \cos x \, dx$ ?
- A.  $-\cos 2x + c$
- B.  $-\frac{1}{2} \cos 2x + c$
- C.  $-\frac{1}{4} \cos 2x + c$
- D.  $\frac{1}{4} \cos 2x + c$
5. What is the domain of  $y = 3 \cos^{-1}(2x + 1)$ ?
- A. Domain  $[-1, 0]$
- B. Domain  $(1, 0)$
- C. Domain  $[-\frac{1}{2}, 0]$
- D. Domain  $[\frac{1}{2}, 0]$

6. In a Mathematics class a teacher can award one of 4 grades, A, B, C or D to each student.  
What is the minimum number of students required so that at least 8 students are guaranteed to receive the same grade?
- A. 28  
B. 29  
C. 32  
D. 33
7. Which of the following is the range of the function  $\frac{1}{x^2 + 1}$
- A.  $(-\infty, \infty)$   
B.  $(-\infty, 1]$   
C.  $(0, 1]$   
D.  $[0, 1]$
8. A curve has an asymptote at  $x = \frac{\pi}{3}$ . Which of the following could be the equation of the curve?
- A.  $y = \sec\left(x - \frac{\pi}{3}\right)$   
B.  $y = \sec\left(x + \frac{\pi}{3}\right)$   
C.  $y = \cot\left(x - \frac{\pi}{3}\right)$   
D.  $y = \operatorname{cosec}\left(x + \frac{\pi}{3}\right)$
9. Which of the following is the primitive of  $\frac{4}{\sqrt{9 - x^2}}$ ?
- A.  $\frac{4}{3} \sin^{-1} \frac{x}{3} + c$   
B.  $\frac{4}{3} \sin^{-1} 3x + c$   
C.  $4 \sin^{-1} 3x + c$   
D.  $4 \sin^{-1} \frac{x}{3} + c$
10. Which of the following best represents the differential equation shown in the slope field?
- A.  $\frac{dy}{dx} = \frac{x}{y} - y^2$   
B.  $\frac{dy}{dx} = \frac{x}{y} + y^2$   
C.  $\frac{dy}{dx} = -\frac{x}{y} - y^2$   
D.  $\frac{dy}{dx} = -\frac{x}{y} + y^2$



**Section II      60 marks****Attempt Questions 11-14****Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available  
For questions in Section II, your response should include relevant mathematical reasoning  
and/or calculations

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**Question 11** (15 marks) Use a SEPARATE writing booklet

- (a) Solve  $\frac{3}{2x-5} \leq -1$  **3**
- (b) Find the derivative of  $\cos^{-1}\left(\frac{3x}{2}\right)$  **1**
- (c) For what values of  $m$  are the distinct vectors  $\begin{pmatrix} m \\ 2m+6 \end{pmatrix}$  and  $\begin{pmatrix} m+1 \\ -1 \end{pmatrix}$  perpendicular? **2**
- (d) Find  $\int \cos^2 4x \, dx$  **2**
- (e) Evaluate  $\int_0^{\frac{\pi}{6}} \cos x \sin^3 x \, dx$  **3**
- (f) For what values of  $m$  is the polynomial  $x^2 - (m+2)x + 2(m+2)$  positive for all  $x$  **2**
- (g) Consider the expansion  $(2x-p)^9$ . The coefficient of  $x^6$  is  $-344064$ . **2**  
Find the value of  $p$

**Question 12** (15 marks) Use a SEPARATE writing booklet

- (a) Find  $\int \frac{dx}{25+4x^2}$  **2**
- (b) Given that  $\sin \theta = \frac{3}{4}$  and  $\frac{\pi}{2} \leq \theta \leq \pi$  determine the exact value of  $\tan 2\theta$  **3**
- (c) (i) Find the derivative of  $x \log_e x - x$  **1**  
(ii) Hence evaluate  $\int_{\sqrt{e}}^e \log_e x \, dx$  **2**
- (d) A mixed volleyball team of eight players is selected from ten males and nine females **2**  
In how many ways can this be done if the team must have at least 2 female players
- (e) The vectors  $\vec{u} = \begin{pmatrix} a \\ 3 \end{pmatrix}$  and  $\vec{v} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$  are parallel. Find the value of  $a$  **2**
- (f) The polynomial  $P(x) = (x-p)^3 + q$  is zero at  $x = 1$  and when divided by  $x$  **3**  
the remainder is  $-7$ . Find the possible values of  $p$

**Question 13** (15 marks) Use a SEPARATE writing booklet

(a) Evaluate  $\int_{-1}^3 x\sqrt{3-x} \, dx$  using the substitution  $u = 3 - x$ . 4

(b) (i) Express  $\sqrt{3} \cos x + \sin x$  in the form  $R \cos(x - \alpha)$  where  $R > 0$  and  $\alpha$  is acute. 2

(ii) Hence solve  $\sqrt{3} \cos x + \sin x = 1$  for  $0 \leq x \leq 2\pi$ . 2

(c) Use the process of Mathematical induction to prove 3

$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$  for all positive integers  $n$ .

(d) A population of mice in a meadow after  $t$  years satisfies the logistic 4

differential equation  $\frac{dP}{dt} = \frac{3P}{2500}(2500 - P)$ , where the initial population of mice is 500.

Given  $\frac{1}{P} + \frac{1}{2500 - P} = \frac{2500}{P(2500 - P)}$ , solve the differential equation to find the

population  $P$  of the mice at time  $t$ . Express your answer in the form

$P = \frac{A}{1 - Be^{-kt}}$  where  $A$ ,  $B$  and  $k$  are integers.

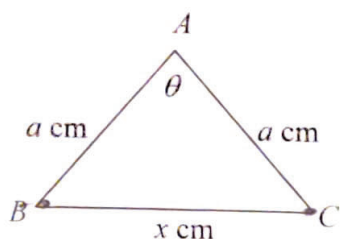
**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

- (a) Solve the differential equation  $(x^2 + 1) \frac{dy}{dx} = 6xy$  where  $x = 1$  and  $y = 2$  giving your answer as  $y$  in terms of  $x$ . 3

- (b) Given that  $y = e^{2x} + e^{-2x}$ , determine the values of constants  $a$  and  $b$  that satisfy the following differential equation  $\frac{d^2y}{dx^2} + a \frac{dy}{dx} + by = 5e^{2x} + e^{-2x}$ . 4

- (c) In the triangle  $ABC$ ,  $AB = AC = a$  cm. The angle  $BAC$  is increasing at the rate of 2 radians/min. Let  $\angle BAC = \theta$  radians and  $BC = x$  cm.



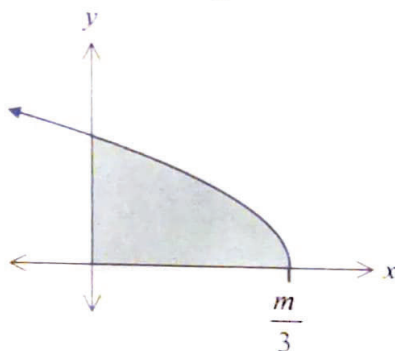
NOT TO  
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- (i) Show that  $x = a\sqrt{2 - 2\cos\theta}$ . 1

- (ii) Show that  $\frac{dx}{d\theta} = \frac{a \sin \theta}{\sqrt{2 - 2\cos \theta}}$ . 1

- (iii) Determine, in terms of  $a$ , the rate with respect to time at which  $BC$  is increasing when  $\theta = \frac{\pi}{3}$  radians. 2

- (d) Let  $f(x) = \sqrt{m - 3x}$  for  $x < \frac{m}{3}$ . The graph of  $y = f(x)$  is shown. 4



The area enclosed by the graph  $y = f(x)$ , the  $x$ -axis and the  $y$ -axis is rotated about the  $y$  axis. Find the value of  $m$  such that the volume of the solid formed is  $\frac{5000\pi}{27}$  cubic units.

**End of Paper**

-1-

MATHEMATICS EXTENSION 1 TRIAL 2022

1.  $\underline{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$   $\underline{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|}$$

$$= \frac{(3 \times -1 + 4 \times 2)}{\sqrt{9+16} \sqrt{1+4}}$$

$$= \frac{5}{5\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$$\theta = \cos^{-1} \frac{1}{\sqrt{5}}$$

$$\approx 63^\circ \quad \textcircled{B}$$

2.  $\textcircled{D} \quad \frac{\sin \theta}{\tan \theta} = \frac{\sin \theta}{\frac{\sin \theta}{\cos \theta}}$

$$= \sin \theta \times \frac{\cos \theta}{\sin \theta}$$

$$= \cos \theta$$

3.  $P(x) = x^3 - x^2 - 5x - 3$

$$P'(x) = 3x^2 - 2x - 5$$

Double root  $\therefore P'(a) = 0$

$$0 = 3a^2 - 2a - 5 \quad \begin{cases} P = -15 \\ S = -2 \\ P = -5.3 \end{cases}$$

$$0 = 3a^2 + 3a - 5a - 5$$

$$0 = 3a(a+1) - 5(a+1)$$

$$0 = (a+1)(3a-5)$$

$$\therefore a = -1 \text{ or } \frac{5}{3}$$

$$P(-1) = (-1)^3 - (-1)^2 - 5(-1) - 3$$

$$= -1 - 1 + 5 - 3$$

$$= 0$$

$$\therefore P(-1) = 0 \quad P'(-1) = 0$$

$$\therefore \textcircled{B}$$

4.  $\int \sin x \cos x \, dx = \frac{1}{2} \int \sin 2x \, dx$

$$= \frac{1}{2} \left( -\frac{\cos 2x}{2} \right) + c$$

$$= -\frac{\cos 2x}{4} + c$$

$\textcircled{C}$

5.  $-1 \leq 2x + 1 \leq 1$

$$-2 \leq 2x \leq 0$$

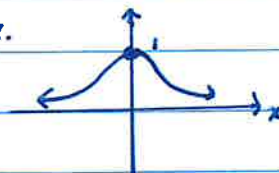
$$-1 \leq x \leq 0$$

$$[-1, 0] \quad \textcircled{A}$$

6.  $4 \times 7 + 1 = 29$

$\textcircled{B}$

7.



$$0 < y \leq 1$$

$$(0, 1] \quad \textcircled{C}$$

8. A.  $y = \sec \theta = \frac{1}{\cos \theta} = 1$

B.  $y = \sec\left(\frac{2\pi}{3}\right)$

C.  $y = \cot(0) = \frac{\cos 0}{\sin 0}$   $\sin 0 = 0 \therefore \text{Undefined at } x=0 \quad \textcircled{C}$

9.  $\int \frac{4}{\sqrt{3^2 - x^2}} \, dx = 4 \sin^{-1} \frac{x}{3} + c \quad \textcircled{D}$

10.  $\textcircled{A} \quad x=1 \quad y=1 \quad \frac{dy}{dx} = 1-1=0 \quad \checkmark$  (A or D)

$$x=-2 \quad y=-1 \quad \frac{dy}{dx} = 1$$

- |    |   |
|----|---|
| 1  | B |
| 2  | D |
| 3  | B |
| 4  | C |
| 5  | A |
| 6  | B |
| 7  | C |
| 8  | C |
| 9  | D |
| 10 | A |

D.  $\frac{dy}{dx} = -2+1 = -1$   $x$

$\therefore \textcircled{A}$

(15 Marks)

11 a)  $\frac{3}{2x-5} \leq -1$

critical values

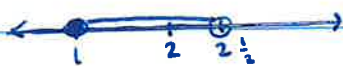
$$x \neq \frac{5}{2}$$

$$\frac{3}{2x-5} = -1$$

$$3 = -2x + 5$$

$$2x = 2$$

$$x = 1$$



test  $x = 2$   $\frac{3}{-1} \leq -1$  ✓

$$1 \leq x < 2\frac{1}{2}$$

[3]

b)  $y = \cos^{-1} \frac{3x}{2}$

$$\frac{dy}{dx} = \frac{-3}{\sqrt{4-9x^2}}$$

$$\begin{cases} a=2 \\ f(x)=3x \\ f'(x)=3 \end{cases}$$

[1]

c) perpendicular when

$$\begin{pmatrix} m \\ 2m+6 \end{pmatrix} \cdot \begin{pmatrix} m+1 \\ -1 \end{pmatrix} = 0$$

$$m(m+1) + -1(2m+6) = 0$$

$$m^2 + m - 2m - 6 = 0$$

$$m^2 - m - 6 = 0$$

$$(m-3)(m+2) = 0$$

$$\therefore m = 3 \text{ or } -2$$

[2]

d)  $\int \cos^2 4x \, dx = \int \left( \frac{1}{2} + \frac{1}{2} \cos 8x \right) dx$

$$= \frac{1}{2}x + \frac{1}{2} \frac{\sin 8x}{8} + c$$

$$= \frac{1}{2}x + \frac{1}{16} \sin 8x + c \quad [2]$$

e)  $\int_0^{\frac{\pi}{6}} \cos x \sin^3 x \, dx$

$$= \left[ \frac{\sin^4 x}{4} \right]_0^{\frac{\pi}{6}}$$

$$= \left( \frac{\sin \frac{\pi}{6}}{4} \right)^4 - \frac{\sin^4 0}{4}$$

$$= \left( \frac{1}{4} \times \frac{1}{2} \right)^4 - 0$$

$$= \frac{1}{64}$$

[3]

f) for positive  $\Delta < 0$ 

$$\Delta = b^2 - 4ac$$

$$= [-(m+2)]^2 - 4(2(m+2))$$

$$= m^2 + 4m + 4 - 8m - 16$$

$$= m^2 - 4m - 12$$

$$= (m-6)(m+2) \quad m = -2, 6$$



$$(m-6)(m+2) < 0 \quad \text{test } m = 0$$

$$-6 \times 2 < 0 \quad \checkmark$$

$$-2 < m < 6$$

[2]

g) coefficient of  $x^6 = {}^9C_6 (2x)^6 (-p)^3$

$$-344064 = -5376p^3$$

$$64 = p^3$$

$$p = 4$$

[2]

(15 Marks)

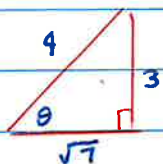
12 a)  $\int \frac{dx}{25+4x^2} = \frac{1}{2} \int \frac{2dx}{5^2+(2x)^2}$

$a=5$   
 $f(x)=2x$   
 $f'(x)=2$

$= \frac{1}{5} \times \frac{1}{2} \tan^{-1} \frac{2x}{5} + c$

$= \frac{1}{10} \tan^{-1} \frac{2x}{5} + c$  [2]

b)  $\sin \theta = \frac{3}{4}$  (Q2)



$\tan \theta = \frac{3}{4}$

$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

$= \frac{2 \times \frac{3}{4}}{1 - (\frac{3}{4})^2}$

$= \frac{-6}{\frac{7}{4}}$

$= -\frac{6 \times 4}{7} = -\frac{24}{7}$

$= \frac{21}{7} \times \frac{4}{7}$

$= \frac{21 \times 4}{7}$

$= 3 \times 4$

[3]

c)

(i)  $y = x \log_e x - x$

$\frac{dy}{dx} = x \times \frac{1}{x} + \log_e x \times 1 - 1$

$= 1 + \log_e x - 1$

$= \log_e x$

[1]

(ii)  $\int_{\sqrt{e}}^e \log_e x \, dx = [x \log_e x - x]_{\sqrt{e}}^e$

$= (e \log_e e - e) - (\sqrt{e} \log_e \sqrt{e} - \sqrt{e})$

$= (e - e) - (\sqrt{e} \log_e e^{\frac{1}{2}} - \sqrt{e})$

$= 0 - (\frac{1}{2} \sqrt{e} \log_e e - \sqrt{e})$

$= 0 - \frac{1}{2} \sqrt{e} + \sqrt{e}$

$= \frac{1}{2} \sqrt{e}$

[2]

d) 8 players 10 males 9 females

At least 2 female = Total - (0F + 1F)

$= {}^{19}C_8 - [{}^9C_0 \times {}^{10}C_8 + {}^9C_1 \times {}^{10}C_7]$

$= 74457$

[2]

2)(e)  $u = \begin{pmatrix} a \\ 3 \end{pmatrix}$   $v = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$   
For parallel  $\begin{pmatrix} a \\ 3 \end{pmatrix} = k \begin{pmatrix} 2 \\ 7 \end{pmatrix}$

$$3 = k \times 7$$

$$\frac{3}{7} = k$$

$$\therefore a = \frac{3}{7} \times 2$$

$$a = \frac{6}{7}$$

[2]

(f)  $P(x) = (x-p)^3 + q$   
 $P(1) = 0$   
 $P(0) = -7$

$$P(1) = (1-p)^3 + q$$

$$0 = (1-p)^3 + q \quad (1)$$

$$P(0) = (-p)^3 + q$$

$$-7 = (-p)^3 + q$$

$$-7 = -p^3 + q$$

$$-7 + p^3 = q \quad (2)$$

Sub (2) into (1)

$$0 = (1-p)^3 - 7 + p^3$$

$$0 = 1 - 3p + 3p^2 - p^3 - 7 + p^3$$

$$0 = 3p^2 - 3p - 6$$

$$0 = p^2 - p - 2$$

$$0 = (p-2)(p+1)$$

$$\therefore p = 2 \text{ or } -1$$

[3]

(13)  $\int_{-1}^3 x \sqrt{3-x} dx$

$$= - \int_4^0 (3-u) u^{\frac{1}{2}} du$$

$$= \int_0^4 (3u^{\frac{1}{2}} - u^{\frac{3}{2}}) du$$

$$= \left[ \frac{3u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{\frac{5}{2}}}{\frac{5}{2}} \right]_0^4$$

$$= \left[ 2u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}} \right]_0^4$$

$$= \left[ 2(4)^{\frac{3}{2}} - \frac{2}{5}(4)^{\frac{5}{2}} \right] - [0-0]$$

$$= 16 - \frac{64}{5}$$

$$= \frac{16}{5}$$

[4]

b)(i)  $\sqrt{3} \cos x + \sin x = R \cos x \cos \alpha + R \sin x \sin \alpha$

$$R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$

$$R = \sqrt{3+1} = 2$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \cos x + \sin x = 2 \cos \left( x - \frac{\pi}{6} \right)$$

[2]

(ii)  $2 \cos \left( x - \frac{\pi}{6} \right) = 1$

$$\cos \left( x - \frac{\pi}{6} \right) = \frac{1}{2}$$

$$\checkmark \text{ ref } \alpha = \frac{\pi}{3}$$

$$\therefore x - \frac{\pi}{6} = \frac{\pi}{3} \text{ or } x - \frac{\pi}{6} = \frac{5\pi}{3}$$

$$x = \frac{\pi}{3} + \frac{\pi}{6}$$

$$x = \frac{5\pi}{3} + \frac{\pi}{6}$$

$$= \frac{\pi}{2} \text{ or } = \frac{11\pi}{6}$$

[2]

(c) step 1: Prove true for  $n=1$

$$\begin{aligned} \text{LHS} &= 2 \times 1! \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 1(1+1)! \\ &= 2! \\ &= 2 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

True for  $n=1$

step 2: Assume true for  $n=k$

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! = k(k+1)! \quad *$$

step 3: Prove true for  $n=k+1$

$$\text{RTP: } \underbrace{2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k!}_{S_k} + [(k+1)^2 + 1](k+1)! = (k+1)(k+2)!$$

$$\begin{aligned} \text{LHS} &= k(k+1)! + [k^2 + 2k + 1 + 1](k+1)! \quad (\text{by assumption}) \\ &= (k+1)! [k + k^2 + 2k + 2] \\ &= (k+1)! [k^2 + 3k + 2] \\ &= (k+1)! (k+2)(k+1) \\ &= \underbrace{(k+1)! (k+2)}_{(k+2)!} (k+1) \\ &= \text{RHS} \end{aligned}$$

step 4: Since true for  $n=1$ , then true for  $n=2, n=3$  and so on for all positive integers.

[3]

$$\text{cd) } \frac{dp}{dt} = \frac{3p}{2500} (2500 - p)$$

Trivial solns:  $p = 0$   
 $p = 2500$

$$\frac{dp}{\frac{p(2500-p)}{2500}} = 3 dt$$

$$\int \frac{2500}{p(2500-p)} dp = \int 3 dt$$

$$\int \left( \frac{1}{p} + \frac{1}{2500-p} \right) dp = 3 \int dt$$

$$\ln |p| - \ln |2500-p| = 3t + C$$

$$\ln \left| \frac{p}{2500-p} \right| = 3t + C$$

$$\left| \frac{p}{2500-p} \right| = e^{3t+C}$$

$$= e^{3t} \times e^C$$

$$= A e^{3t} \quad \text{where } A = e^C$$

$$\frac{p}{2500-p} = A e^{3t} \quad A = \pm e^C$$

when  $t = 0$   
 $p = 500$

$$\frac{500}{2000} = A e^0$$

$$\frac{1}{4} = A$$

$$\therefore \frac{p}{2500-p} = \frac{1}{4} e^{3t}$$

$$4p = e^{3t} (2500 - p)$$

$$4p = 2500 e^{3t} - p e^{3t}$$

$$4p + p e^{3t} = 2500 e^{3t}$$

$$p(4 + e^{3t}) = 2500 e^{3t}$$

$$p = \frac{2500 e^{3t}}{4 + e^{3t}} \quad \div e^{3t}$$

$$= \frac{2500}{\frac{4}{e^{3t}} + 1}$$

$$= \frac{2500}{1 + 4e^{-3t}}$$

$$= \frac{2500}{1 + 4e^{-3t}}$$

$$B = -4$$

[4]

or  $p = \frac{2500}{1 - (-4)e^{-3t}}$

(15 Marks)

-7-

14  $(x^2+1) \frac{dy}{dx} = 6xy$

a)  $\int \frac{dy}{y} = \int \frac{6x}{x^2+1} dx$

$$\ln |y| = 3 \int \frac{2x}{x^2+1} dx$$

$$\ln |y| = 3 \ln |x^2+1| + c$$

$$x=1, y=2 \quad \ln 2 = 3 \ln 2 + c$$

$$-2 \ln 2 = c$$

$$\therefore \ln y = 3 \ln (x^2+1) - 2 \ln 2$$

$$\ln y = \ln (x^2+1)^3 - \ln 2^2$$

$$y = \frac{(x^2+1)^3}{4}$$

[3]

b)  $y = e^{2x} + e^{-2x}$

$$\frac{dy}{dx} = 2e^{2x} - 2e^{-2x}$$

$$\frac{d^2y}{dx^2} = 4e^{2x} + 4e^{-2x}$$

$$\begin{aligned} \frac{d^2y}{dx^2} + a \frac{dy}{dx} + by &= 4e^{2x} + 4e^{-2x} + 2ae^{2x} - 2ae^{-2x} + be^{2x} + be^{-2x} \\ &= (4+2a+b)e^{2x} + (4-2a+b)e^{-2x} \end{aligned}$$

Equating coefficients:

$$4 + 2a + b = 5$$

$$4 - 2a + b = 1$$

$$2a + b = 1 \quad (1)$$

$$-2a + b = -3 \quad (2)$$

$$+ \quad -2a + b = -3 \quad (2)$$

$$-2a - 1 = -3$$

$$2b = -2$$

$$-2a = -2$$

$$b = -1$$

$$a = 1$$

$$\therefore \underline{a=1 \quad b=-1}$$

[4]

(i) cosine rule:

$$c^2 = a^2 + b^2 - 2ab \cos c$$

(ii)  $x^2 = a^2 + a^2 - 2a^2 \cos \theta$

$$= 2a^2 - 2a^2 \cos \theta \quad (\text{factorise } a^2)$$

$$\underline{x = a \sqrt{2 - 2 \cos \theta}} \quad \text{which is required} \quad [1]$$

ii)  $\frac{dx}{d\theta} = a \times \frac{1}{2} (2 - 2 \cos \theta)^{-\frac{1}{2}} \times 2 \sin \theta$

$$= \frac{a \sin \theta}{\sqrt{2 - 2 \cos \theta}}$$

which is required

[1]

(iii)  $\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$

$$= \frac{a \sin \theta}{\sqrt{2 - 2 \cos \theta}} \times 2 \quad \text{when } \theta = \frac{\pi}{3}$$

$$= \frac{2a \sin \frac{\pi}{3}}{\sqrt{2 - 2 \cos \frac{\pi}{3}}}$$

$$= \frac{2a \frac{\sqrt{3}}{2}}{\sqrt{1}}$$

$$= \underline{\underline{\sqrt{3}a}} \text{ cm/min}$$

[2]

d)  $V = \pi \int_0^m x^2 dy$

$$= \pi \int_0^m (y^4 - 2my^2 + m^2) dy$$

$$= \frac{\pi}{9} \left[ \frac{y^5}{5} - 2m \frac{y^3}{3} + m^2 y \right]_0^m$$

$$= \frac{\pi}{9} \left[ \frac{(\sqrt{m})^5}{5} - \frac{2m}{3} (\sqrt{m})^3 + m^2 \sqrt{m} \right]$$

$$= \frac{\pi}{9} \left[ \frac{m^2 \sqrt{m}}{5} - \frac{2}{3} m^2 \sqrt{m} + m^2 \sqrt{m} \right]$$

$$= \frac{\pi}{9} \times m^2 \sqrt{m} \left( \frac{1}{5} - \frac{2}{3} + 1 \right)$$

$$= m^2 \sqrt{m} \frac{\pi}{9} \times \frac{8}{15}$$

$$= \underline{\underline{\frac{8}{135} m^2 \sqrt{m} \pi}}$$

$$y = \sqrt{m-3x} \quad x=0 \quad y=\sqrt{m}$$

$$y^2 = m-3x$$

$$3x = m-y^2$$

$$x = \frac{m-y^2}{3}$$

$$x^2 = \frac{(m-y^2)^2}{9}$$

$$= \frac{m^2 - 2my^2 + y^4}{9}$$

$$\frac{8}{135} m^{5/2} \pi = \frac{5000\pi}{27}$$

$$m^{5/2} = 3125$$

$$\underline{\underline{m = 25}}$$

[4]